

Philosophy of Science
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Lecture 2

Knowledge, Truth, and Facts

Motivation of this lecture

- In International Studies, you strive daily to acquire knowledge
 - Chiefly in the form of propositions
- But in what conditions may we say...
 - "I know that P "?
 - "Proposition P is true"?
 - "It is a fact that P "?
- In this lecture, you learn to analyse the concepts of knowledge, truth, and fact

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Overview of this lecture

- Classification of forms of knowledge
 - Focus on propositional knowledge
- What is knowledge?
 - Knowledge as justified true belief
- What is truth?
 - Correspondence theory of truth
- What is a fact?
 - Permanence of facts and of truths

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Reading for this lecture

- Duncan Pritchard, *What Is This Thing Called Knowledge?*
 - Third edition
 - Routledge, 2014
 - Read chapter 1, "Some Preliminaries", pp. 3–9
- Available on Brightspace

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Three forms of knowledge

- Knowledge by acquaintance
 - I know The Hague
- Know-how: practical knowledge or skill
 - I know how to ride a bicycle
- Knowing that: propositional knowledge
 - I know that P
 - ... where P is a proposition
 - Propositional knowledge is our focus today

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Propositional knowledge

- We regard propositional knowledge as the highest form of knowledge. Why?
 - It is knowledge of facts
 - Important in science, in logical reasoning, in arguments
 - Fits in the intellectualist Cartesian tradition
- Each of us may claim many items of propositional knowledge...
 - in various categories
 - Some examples follow

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Knowledge that P : examples

- Knowledge of observed states of affairs
 - I know that there is a door over there
- Knowledge of future events
 - I know that the sun will rise tomorrow
- Knowledge of conceptual relations
 - I know that red is a colour
- Knowledge of mathematical facts
 - I know that $7 + 5 = 12$

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What is knowledge?

- The JTB account of knowledge offers an answer to this question
 - It says that propositional knowledge is justified true belief
 - Plato proposed this idea in his dialogue, *Theaetetus*, around 370 BCE
- The JTB account analyses the statement "A knows that P "
 - A is a person; P is a proposition

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K is JTB

- *A* knows that *P* if and only if:
 1. *P* is true
 2. *A* believes that *P*
 3. *A* is justified in believing that *P*
- This is the JTB account of knowledge
 - If conditions 1–3 hold, then *A* knows that *P*
 - If not, then this is not a case of knowledge

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JTB account: example

- Jo knows that it is raining if and only if:
 1. "It is raining" is true—i.e., it is raining
 2. Jo believes that it is raining
 3. Jo is justified in believing that it is raining
- These conditions specify the meaning of "Jo knows that it is raining"
 - ... and the circumstances in which we may legitimately assert this

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What does the JTB account do?

- The JTB account of knowledge...
 - is a good example of conceptual analysis in philosophy
 - unpacks the concept "knowledge", showing how it relates to three more basic concepts: "truth", "belief", and "justification"
 - fixes what it means to say "*A* knows that *P*"
 - specifies what conditions must be fulfilled for "*A* knows that *P*" to hold

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JTB account: condition 1

- "*P* is true"
 - Condition 1 hinges on whether *P* is true
 - Not on whether we are aware that it is true or any other attainment
- Because of condition 1, "*A* knows that *P*" logically entails "*P* is true"
 - No-one can know that *P* if *P* is not true
 - Example: one may *believe* that Milan is the capital of Italy, but no-one can *know* this

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Condition 1: consequence

- Everyone may have their own beliefs
 - It is possible that *A* believes that *P* and *B* believes that not-*P*
- But knowledge is the same for all
 - It is possible that *A* knows that *P* and *B* does not
 - But it is not possible that *A* knows that *P* and *B* knows that not-*P*
- Belief is private; knowledge is public

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Condition 1: discussion

- Beginner's frequent question
 - "But how can we be sure that *P* is true?"
- A pressing question in science!
 - Answers involve empirical data, textual evidence, proof, testimony, etc.
- ... but not a worry for the JTB account
 - Task of JTB is to specify how "knowledge" relates to "truth" and other concepts
 - Not to solve all problems of science

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JTB account: condition 2

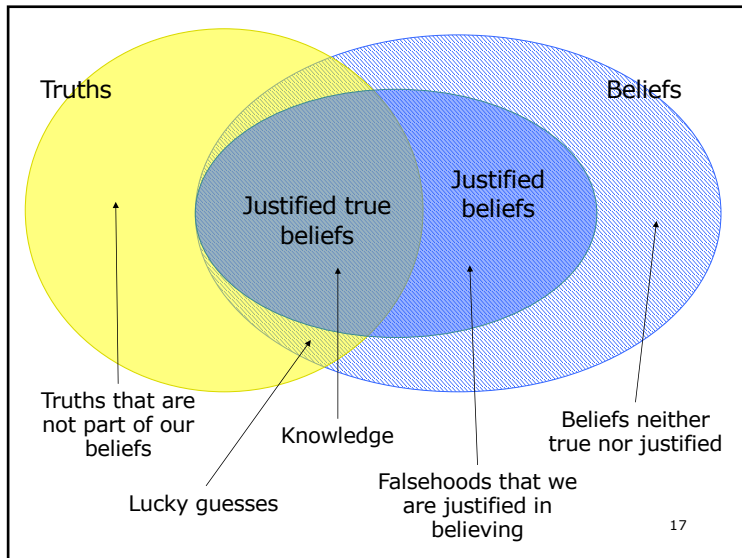
- "A believes that *P*"
 - Establishes that knowledge is, in part, a psychological phenomenon
 - To know is to be in a particular mental state
- Implications:
 - Belief is a precondition for knowledge
 - "I know that *P*, but I do not believe that *P*" is self-contradictory

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JTB account: condition 3

- "A is justified in believing that *P*"
 - To be justified: to have good grounds for holding a belief
 - Purpose of condition 3: to rule out lucky guesses as cases of knowledge
- Justification vs. truth
 - Justification depends on available evidence and argument: it rises and falls with time
 - Justified beliefs can be false—as the history of science demonstrates

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From knowledge to truth

- The JTB account...
 - analyses the concept “knowledge” in terms of “truth”, “belief”, and “justification”
- But what does “truth” consist in?
 - Let us now investigate!

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What is truth?

- Truth is a property only of propositions
 - ... and of systems of propositions, such as scientific theories
 - Other knowledge elements have success predicates of their own—e.g., concepts are adequate; arguments are valid; etc.
- But which property is it?
 - Best answer to this question: the “correspondence theory” of truth
 - Truth amounts to correspondence with facts

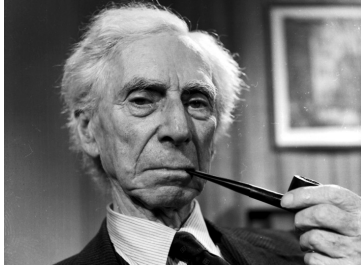
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Correspondence theory of truth

- A proposition is true if and only if there is a fact corresponding to it
 - Facts make our utterances true or false
 - Whether a proposition is true depends on no other factor
- Example:
 - The proposition, “Snow is white”, is true if and only if snow is white
- The most influential account of truth
 - Bertrand Russell advocated it in the 1910s

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Bertrand Russell, 1872–1970



- British logician, philosopher, political activist
- One of the founders of analytic philosophy
- Jailed for pacifism during World War I

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Russell on truth

Thus a belief is true when there is a corresponding fact, and is false when there is no corresponding fact.

It will be seen that minds do not create truth or falsehood. They create beliefs, but when once the beliefs are created, the mind cannot make them true or false What makes a belief true is a *fact*, and this fact does not (except in exceptional cases) in any way involve the mind of the person who has the belief.

—Bertrand Russell, *The Problems of Philosophy*, 1912, chapter 12

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What is a fact?

- A fact is a state of affairs
 - It makes a proposition true, according to the correspondence theory
 - Example: the fact of the whiteness of snow makes the proposition “Snow is white” true
- This means that...
 - facts are something in the world, outside the realm of propositions

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Facts are not propositions

- Propositions are truth bearers
 - A proposition can state or convey a truth
- Facts are truthmakers
 - A fact makes a proposition true
- This is why...
 - facts are not true or false: truth and falsity are properties only of propositions
 - We say of facts just that they are or are not

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Truth and facts: a metaphor

- . Astronomers study the planets
 - . Mercury, Venus, Mars, etc.
 - . Astronomers want their claims about the planets to be true rather than false
 - . What makes astronomers' claims about the planets true or false? The planets do
- . Planets themselves are not true or false
 - . They simply are or are not in the world
 - . My advice: think of facts like planets

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Permanence of facts

- . Philosophers think of facts as permanent and eternally stable
 - . Facts do not "change in time"
 - . Though they may be keyed to particular times
- . Example: it is a fact that...
 - . France in 1680 is a monarchy
 - . France in 2026 is a republic
- . These facts will never change

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Consequences for truths 1

- . If P is true...
 - . then P always has been and will be true
 - . To make this clear, however, it is necessary to formulate P explicitly and fully
- . Example:
 - . "France is a monarchy" might seem true at some times and false at other times
 - . But "France is a monarchy in 1680" is true regardless of time

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Consequences for truths 2

- . Our estimate of the truth of a proposition P changes in time, of course
 - . Once, humans believed that dolphins were fish
 - . Now we believe that they are mammals
- . But if the proposition "Dolphins are mammals" is true...
 - . then it has always been true
 - . Only, we have not always believed it

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Some complications

- The Gettier problem revealed a flaw
 - The JTB account lets through some cases that we intuitively reject as knowledge
 - Philosophers have discussed adding a fourth criterion to close the gap: JTB+
- Alternative conceptions of truth and of facts are available. Examples:
 - Coherence theory of truth
 - Constructivist accounts of facts
- These go beyond our present scope

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Recapitulation

- JTB account of knowledge
 - A knows that P if and only if: P is true; A believes that P ; A is justified in believing that P
- Correspondence theory of truth
 - A proposition P is true if and only if there is a fact corresponding to it
- Facts: what make propositions true
 - Facts do not change in time

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